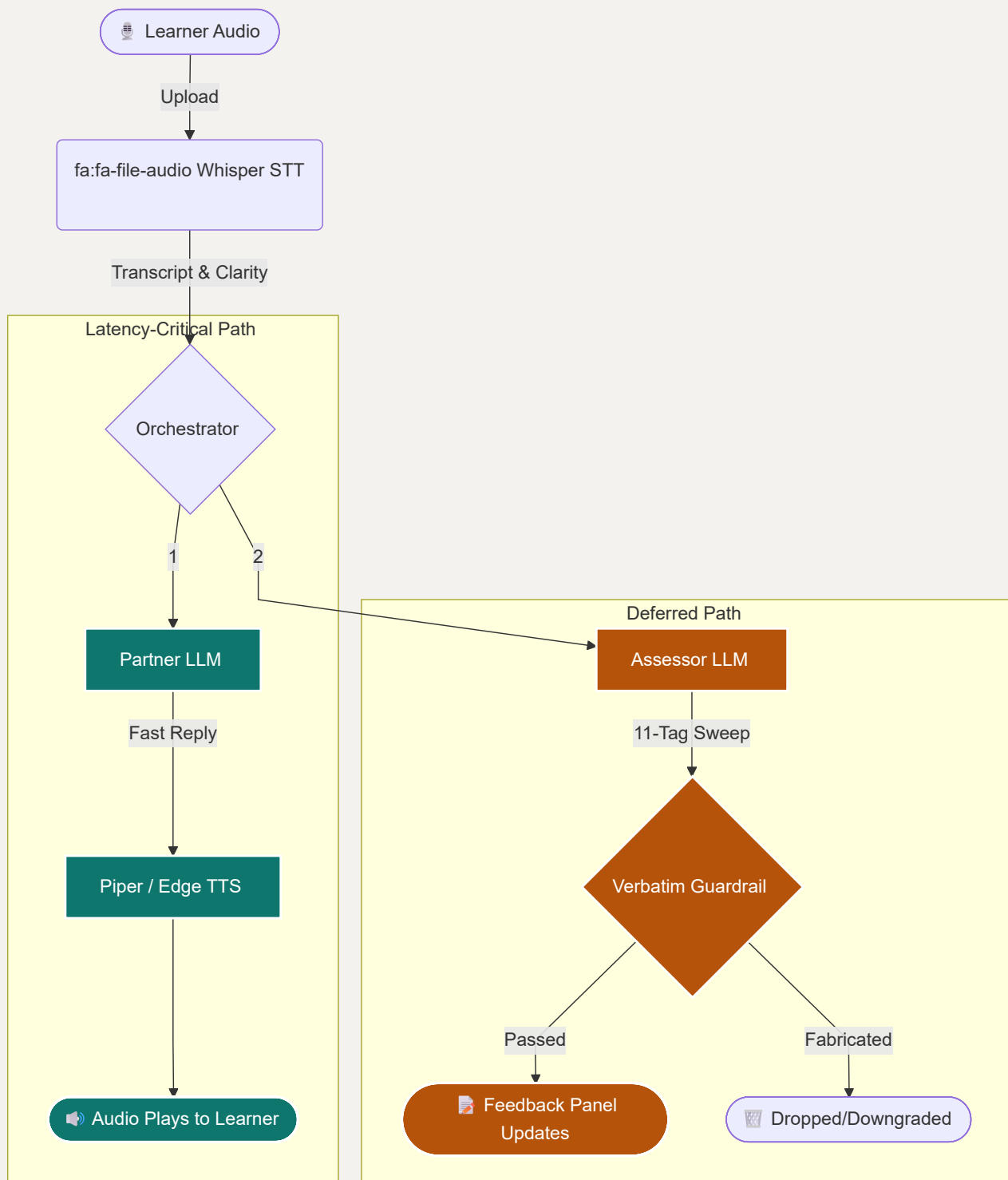


CEFR Compound AI: Architecture & Operations Documentation

Below is the state of the CEFR Compound AI system. The architecture has moved from a conceptual prototype to a latency-hiding, mathematically rigorous pipeline[cite: 15, 18].

1. System Architecture & Turn Orchestration

The core innovation of this revision is **Deferred Assessment**[cite: 14]. By splitting the conversational response from the grammatical assessment, the system hides the heavy language model prefill penalty and returns audio to the learner almost instantly[cite: 15].



✓ 2. The Good: Architectural Masterstrokes

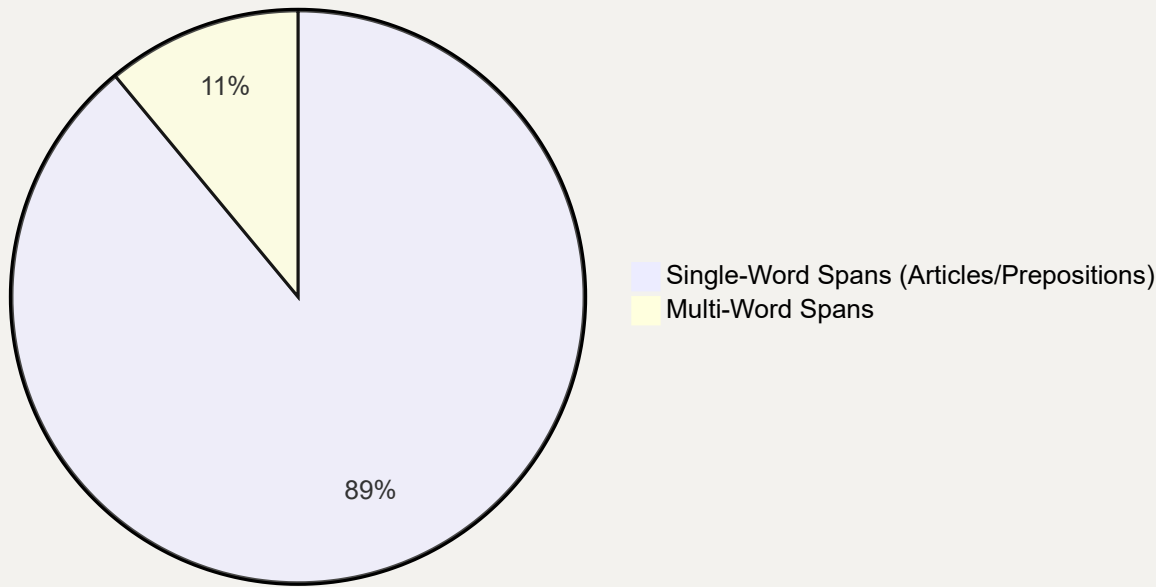
The engineering foundation is now defensively designed against LLM hallucinations, UI crashes, and cloud provider outages[cite: 15, 18, 21].

COMPONENT	MECHANISM	IMPACT
Prefix-Cache Prompting	<code>ASSESSOR_PROMPT_VARIANT=cached</code> hoists the rules, taxonomy, and guardrails into a byte-stable system message[cite: 14, 18].	Guarantees KV cache hits on local runtimes. Drops per-turn prefill from ~725 tokens to ~95 tokens[cite: 14, 18].
Deferred Assessment	<code>DEFER_ASSESSMENT=true</code> removes the assessor from the conversational critical path[cite: 14, 18].	A 40-second local assessment costs the active conversation nothing[cite: 15]. Improves pedagogy by delaying corrections[cite: 15].
Verbatim Guardrails	The <code>_substring_match</code> code forces the LLM to quote the learner character-for-character[cite: 7, 15].	If the model fabricates a quote, it is instantly discarded or downgraded before reaching the UI[cite: 15].
Resilient TTS	<code>TTS_BACKEND=auto</code> swaps out the fragile Microsoft <code>edge-tts</code> websocket for local Piper TTS[cite: 14, 18].	Synthesizes a 4-second reply in under a second natively on CPU, preserving CEFR-specific tempos via <code>length_scale</code> [cite: 17, 18, 20].
Paced Benchmarking	The <code>eval_assessor.py</code> harness uses a token-bucket rate limiter (<code>--rpm 25</code>) and on-disk prompt caching (<code>--cache</code>) [cite: 8, 18].	919-case benchmarks run safely under free-tier API limits. Rerunning unchanged prompts costs zero API calls[cite: 8, 17, 18].
UI Asynchrony	Gradio yields an async generator while grading, and catches <code>CancelledError</code> [cite: 15, 16, 19].	Displays "Checking your English..." instead of falsely claiming a clean sentence[cite: 15, 16]. Client disconnects gracefully degrade the panel instead of crashing[cite: 15, 19].
Drift Guard	<code>check_tree.py</code> serves as a manifest wired to a Git pre-commit hook[cite: 18, 21].	Instantly fails the build if stray agents or Docker volumes resurrect undeclared code like <code>eval/benchmark.py</code> [cite: 21].

🚧 3. The Hurdles: What Still Requires Measurement

Engineering cannot fix physical and acoustic realities. The following items require empirical measurement, not code modifications[cite: 10, 17].

ble Errors in Audited Corpus (Cambridge W&I)"



OPERATIONAL HURDLE	THE REALITY	REQUIRED ACTION
STT Recall Ceiling	Whisper silently "repairs" broken grammar (e.g., transcribing "I have went" as "I have gone"), making the assessor blind to the error[cite: 10].	Run 50 real student audio recordings through the 3-pass protocol in <code>tools/make_spoken_gold.py</code> [cite: 10, 17]. If ceiling is < 80%, STT is the binding constraint[cite: 10].
The Domain Illusion	The Cambridge W&I benchmark (<code>gold_clean.jsonl</code>) tests written essays. It lacks stutters, false starts, and conversational length[cite: 8, 10].	Acknowledge these metrics as a capability floor, not the product's actual spoken performance proxy[cite: 8].

OPERATIONAL HURDLE	THE REALITY	REQUIRED ACTION
SLM Hardware Limits	11-way classification and verbatim extraction is complex for a 7B local Small Language Model[cite: 17].	Gate local deployments on rigorous A/B testing against the cloud baseline; only ship if confidence intervals overlap[cite: 17].
Cloud-Bound Partner	The conversational partner sits on the strict latency-critical path and requires frontier-model quality[cite: 17].	Do not move offline. Moving it trades the most visible quality surface for minimal savings on the cheapest API call[cite: 17].